

Insurance Policy Management Database

Designed for: Spider-Man Insurance Agency

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Course: Data Management and SQL

FORDHAM UNIVERSITY

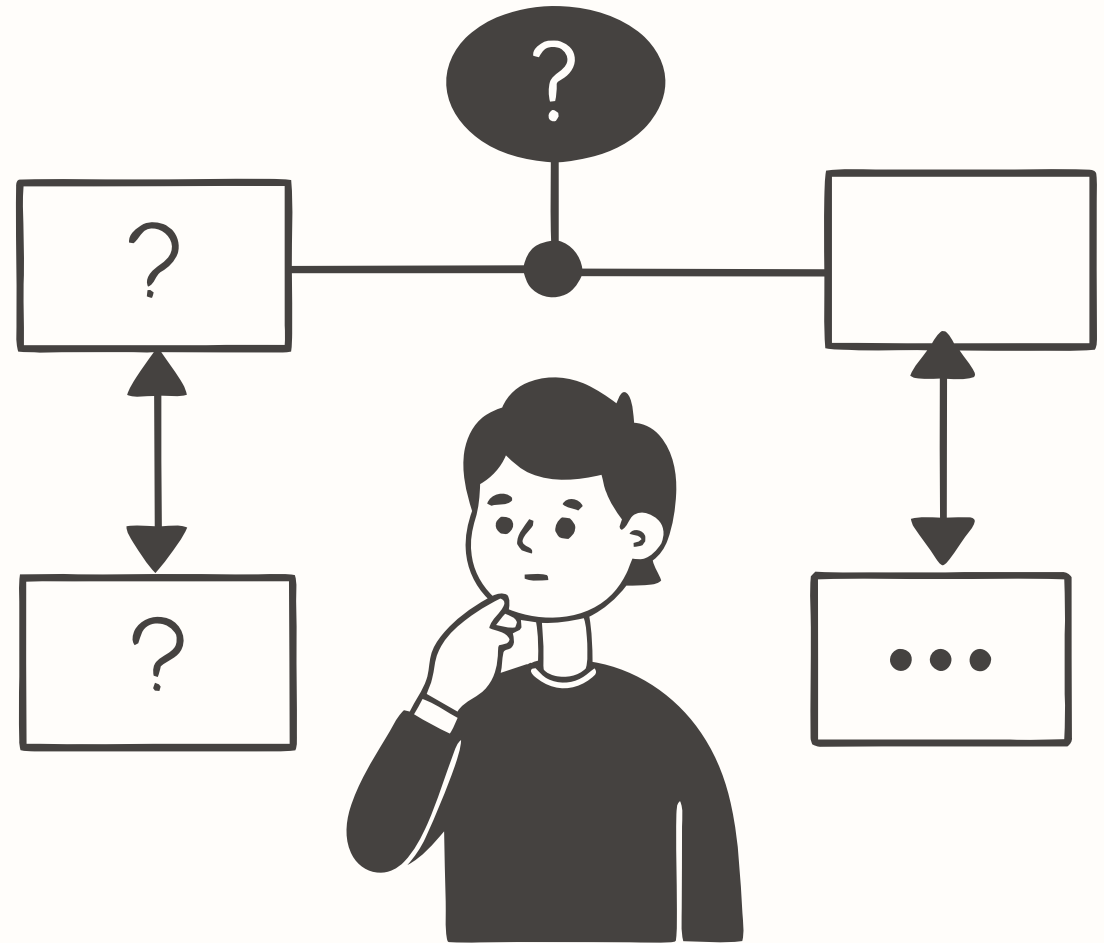
The Business Problem

Insurance agencies manage many moving parts:

- Clients and agents
- Auto and life policies
- Claims, invoices, and payments
- Multiple insurance companies

Without a centralized system:

- Information gets scattered
- Reporting is slow
- Performance is hard to track
- Important decisions rely on incomplete data





The Solution

We built a single, integrated database that gives managers a clear picture of:

- All client and policy information
- Which agents manage which clients
- Policy revenue and payments
- Claims activity
- Invoice and payment history

This system improves accuracy, efficiency, and helps business owners make smarter decisions.

Business Workflow Overview

How the business operates:

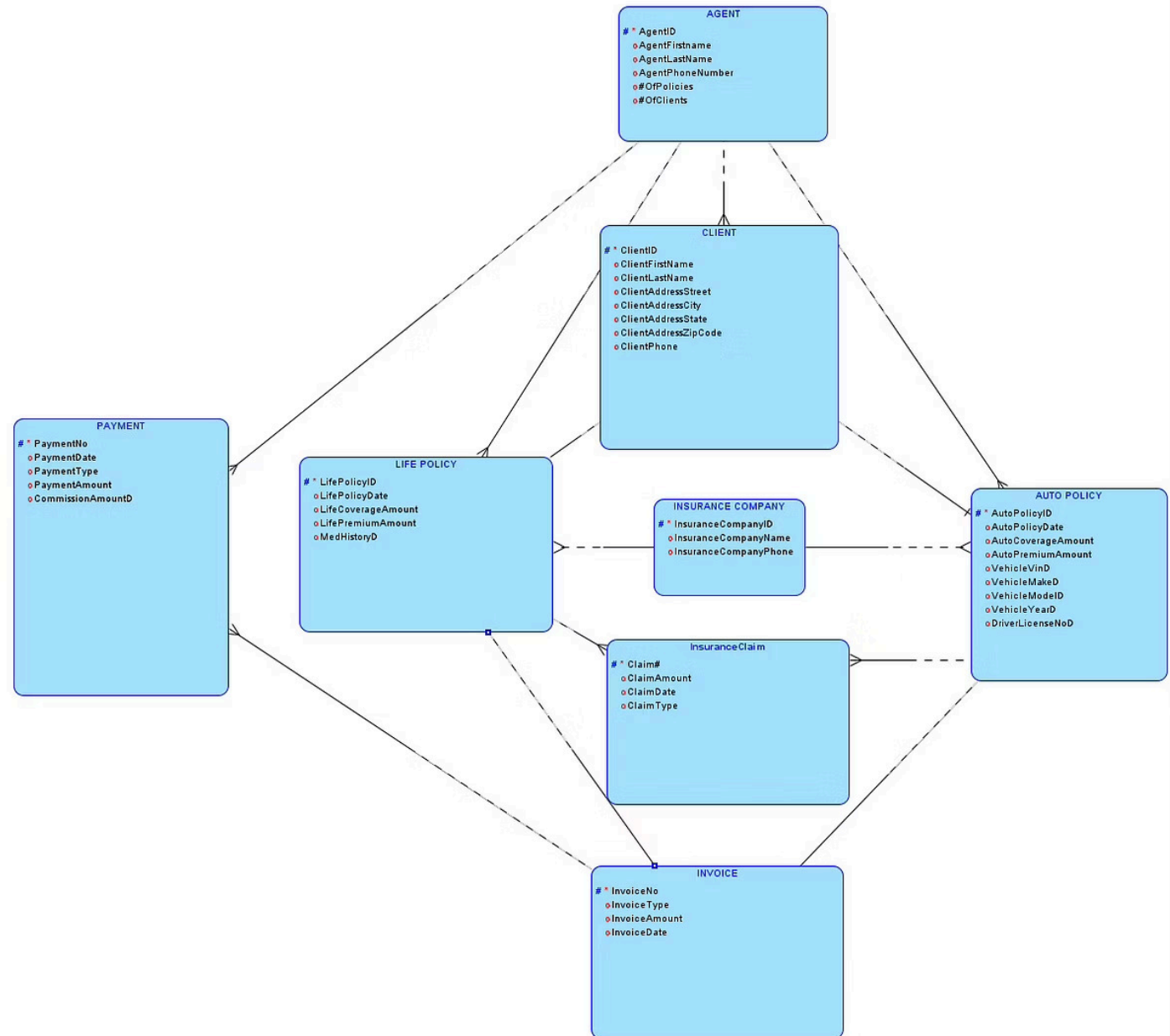


Entity Relationship Diagram (ERD)

The ERD shows how the key parts of the business connect:

- Client
- Agent
- Auto policy
- Life policy
- Vehicle
- Claim
- Invoice
- Payment
- Insurance company

This is the foundation for all reporting and analysis.



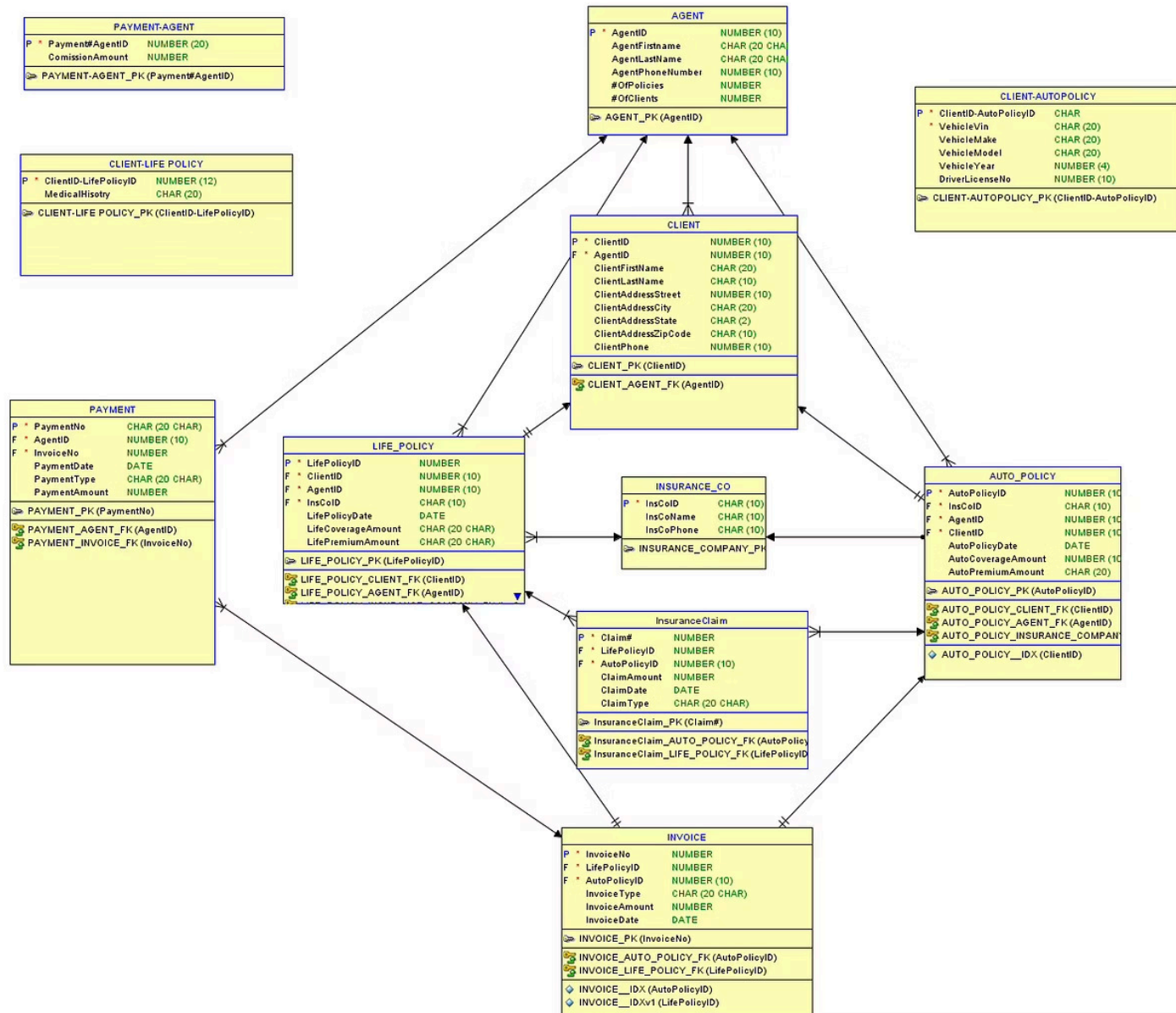
Relational Model

We converted the ERD into actual database tables. Each table represents one part of the business, such as:

- CLIENT
- AGENT
- AUTO_POLICY
- LIFE_POLICY
- INSURANCE_COMPANY
- INVOICE
- PAYMENT

(And the relation tables)

This structure prevents duplication and ensures data integrity.

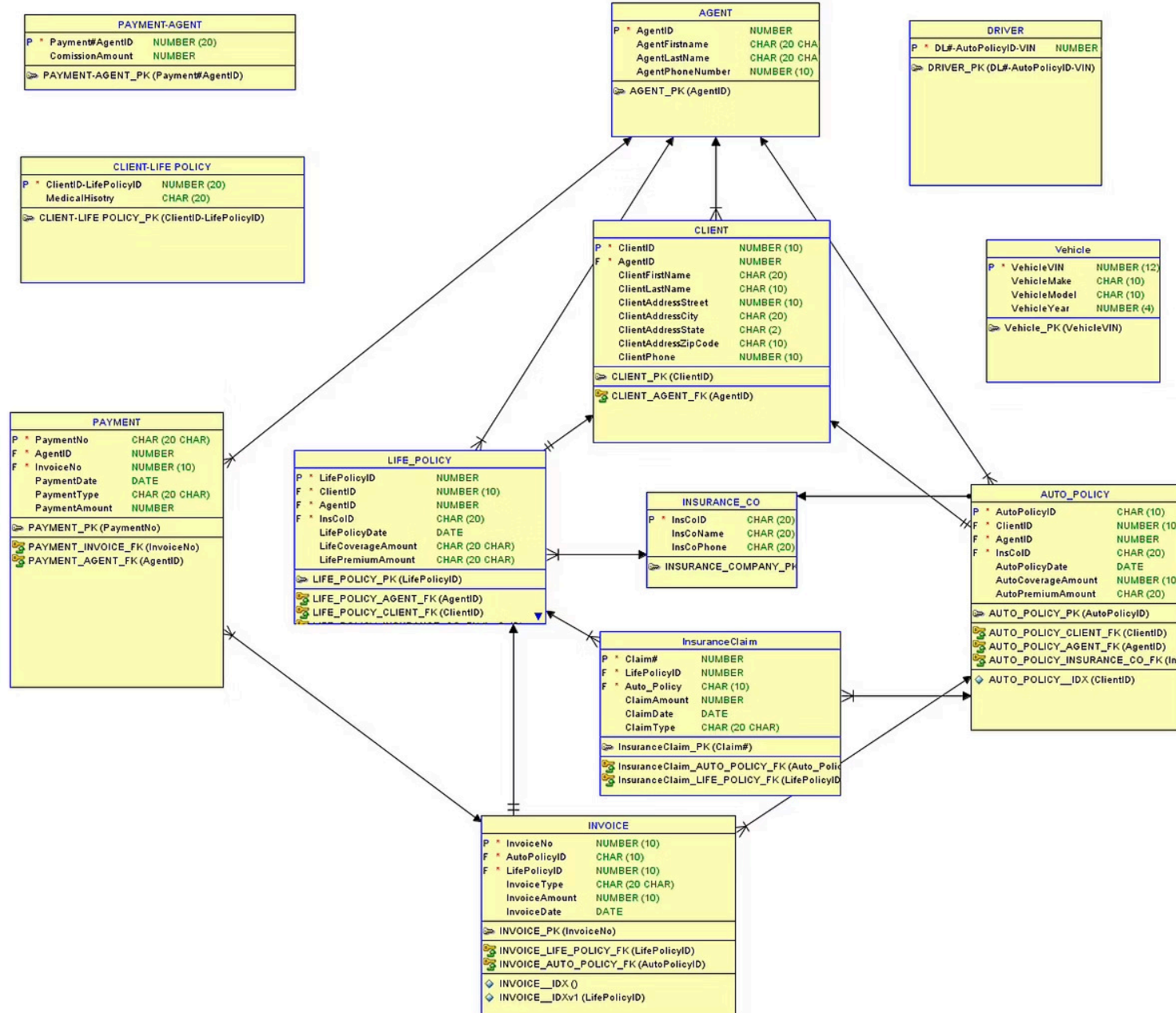


Normalized Relational Model

We refined the design so each table contains only the information it should:

- Removed duplicate or derived attributes
- Moved relationship-specific details into relationship tables (DRIVER, VEHICLE, etc.)
- Ensured clean 3rd Normal Form for accuracy and efficiency

The result is a reliable, scalable database.



Surprise Demo: Top-Performing Agents

This query shows which agents bring in the highest total auto-policy premiums:

AGENT 187
Gavin
TOTAL PREMIUM: 300

AGENT 262
Marcellus
TOTAL PREMIUM: 200

AGENT 879
Luca
TOTAL PREMIUM: 120

```
1  SELECT
2
3      a.AgentID,
4
5      a.AgentFirstName,
6
7      SUM(ap.AutoPremiumAmount) AS TotalPremium
8
9  FROM AGENT a
10
11 JOIN AUTO_POLICY ap
12
13     ON a.AgentID = ap.AgentID
14
15 GROUP BY
16
17     a.AgentID,
18
19     a.AgentFirstName
20
21 ORDER BY
22
23     TotalPremium DESC;
```

Helps with performance reviews, bonuses, hiring, and scheduling.

Closing Summary

What our database delivers:

Centralized business data

Accurate reporting

Clear view of revenue and
payments

Streamlined policy and claim tracking

Most importantly, improved decision-
making for managers

THANK YOU!